

EBS VOLUME

- An **EBS volume** (Elastic Block Store volume) in **AWS** is a block-level storage solution designed for use with Amazon EC2 instances.
- EBS volumes provide persistent storage, meaning they retain data even if the EC2 instance is stopped or restarted.

Key Features of EBS Volumes

1. **Persistence:** EBS volumes remain available independently of the lifecycle of an EC2 instance.
2. **Performance Types:**
 - **General Purpose SSD (gp2, gp3):** Suitable for most workloads with balanced performance.
 - **Provisioned IOPS SSD (io1, io2):** For I/O-intensive applications needing high IOPS, like databases.
 - **Throughput Optimized HDD (st1):** Ideal for large, sequential workloads like log processing.
 - **Cold HDD (sc1):** Cost-effective storage for infrequently accessed data.
3. **Scalability:** EBS volumes can be scaled in size, IOPS, or throughput as requirements grow.
4. **Snapshots:** You can take incremental backups (snapshots) of EBS volumes to Amazon S3, which can be used to restore data or create new volumes.
5. **Encryption:** EBS volumes support encryption for data at rest, in-transit, and for snapshots.
6. **Availability & Durability:** Each EBS volume is automatically replicated within its Availability Zone to protect from hardware failure, ensuring high availability.

Created an ec2 instance

Instances (1) [Info](#)

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

All states ▼

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☐	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public
☐	ebs	i-06ecf5d9fff20cbde	Running	t2.micro	2/2 checks passed View alarms +		us-east-1a	ec2-3-8

Creating and Using an EBS Volume

1. **Create the Volume:** Specify the size, volume type, and Availability Zone.
2. **Attach to an EC2 Instance:** After creation, attach it to a running or stopped instance.
3. **Format and Mount:** Once attached, format it with a filesystem and mount it if needed.
4. **Snapshotting:** You can periodically take snapshots for backups.
5. **Detach:** Detach the volume if no longer needed but do not delete if you want to preserve data.

Created an additional EBS volume for the ec2 instance and attached it

Successfully attached volume [vol-0b09d0ef176fcab56](#) to instance [i-06ecf5d9fff20cbde](#).

Volumes (1/2) [Info](#)

Search

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	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
☒	-	vol-0b09d0ef176fcab56	gp3	2 GiB	3000	125	-	2024/11/11 15:29 GMT+5:..
☐	-	vol-0b6801bafcd60f06e	gp2	8 GiB	100	-	snap-0682a7b...	2024/11/11 15:15 GMT+5:..

Volume ID: [vol-0b09d0ef176fcab56](#)

Details

Status checks

Monitoring

Tags

Volume ID

[vol-0b09d0ef176fcab56](#)

AWS Compute Optimizer finding

❗ Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Fast snapshot restored

No

Size

2 GiB

Volume state

✔ In-use

Availability Zone

us-east-1a

Type

gp3

IOPS

3000

Created

[Mon Nov 11 2024 15:29:01 GMT+0530 \(India Standard Time\)](#)

Volume status

⋮ Insufficient data

Throughput

125

Multi-Attach enabled

No

Logged into the server to mount the volume and create the test.file

```
ec2-user@ip-172-31-30-251:~  
[ec2-user@ip-172-31-30-251 ~]$ lsblk  
NAME        MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINT  
xvda        202:0    0   8G  0 disk  
└─xvda1     202:1    0   8G  0 part /  
xvdh        202:112  0    2G  0 disk  
[ec2-user@ip-172-31-30-251 ~]$ sudo file -s /dev/xvdh  
/dev/xvdh: data  
[ec2-user@ip-172-31-30-251 ~]$ sudo lsblk -f  
NAME        FSTYPE LABEL UUID                                MOUNTPOINT  
xvda  
└─xvda1     xfs      /        6c60627f-646c-4c34-8719-b99f46f2b03f /  
xvdh  
[ec2-user@ip-172-31-30-251 ~]$ sudo mkfs -t xfs /dev/xvdh  
meta-data=/dev/xvdh            isize=512    agcount=4, agsize=131072 blks  
                =               sectsz=512    attr=2, projid32bit=1  
                =               crc=1        finobt=1, sparse=1, rmapbt=0  
                =               reflink=1     bigtime=0 inobtcount=0  
data        =               bsize=4096    blocks=524288, imaxpct=25  
                =               sunit=0      swidth=0 blks  
naming      =version 2        bsize=4096    ascii-ci=0, ftype=1  
log         =internal log    bsize=4096    blocks=2560, version=2  
                =               sectsz=512    sunit=0 blks, lazy-count=1  
realtime    =none           extsz=4096    blocks=0, rtextents=0  
[ec2-user@ip-172-31-30-251 ~]$ sudo mkdir /data  
[ec2-user@ip-172-31-30-251 ~]$ sudo mount /dev/xvdh /data  
[ec2-user@ip-172-31-30-251 ~]$  
  
[ec2-user@ip-172-31-30-251 data]$ ls  
myebs  test.file  
[ec2-user@ip-172-31-30-251 data]$ sudo chmod 777 test.file  
[ec2-user@ip-172-31-30-251 data]$ vi test.file  
[ec2-user@ip-172-31-30-251 data]$ sudo cat test.file  
successfully mounted my ebs volume!  
[ec2-user@ip-172-31-30-251 data]$
```

Creating snapshot from ebs volume(2gp)

Snapshots (1/1) Info

Owned by me

Search

Refresh

Recycle Bin

Actions

Create snapshot

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Settings

☒	Name	Snapshot ID	Volume size	Description	Storage tier	Snapshot status	Started
☒	volume to ...	snap-0c593d8da5a7fea82	2 GiB	mynsop	Standard	Completed	2024/11/11 16:45

Snapshot ID: snap-0c593d8da5a7fea82 (volume to snapshot)

Details

Snapshot settings

Storage tier

Tags

Snapshot ID

snap-0c593d8da5a7fea82 (volume to snapshot)

Started

Mon Nov 11 2024 16:45:38 GMT+0530 (India Standard Time)

Source volume

Volume ID

vol-0b09d0ef176fcab56

Progress

Available (100%)

Product codes

-

Volume size

2 GiB

Snapshot status

Completed

Fast snapshot restore

-

Owner

600627341010

Description

mynsop

4gp Volume was created using a snapshot

Volumes (1/3) Info

Search

Refresh

Actions

Create volume

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Settings

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
☐	-	vol-0b09d0ef176fcab56	gp3	2 GiB	3000	125	-	2024/11/11 15:29 GMT+5:...
☒	-	vol-04d3eeb483ab7199d	gp3	4 GiB	3000	125	snap-0c593d8...	2024/11/11 16:53 GMT+5:...
☐	-	vol-0b6801bafcd60f06e	gp2	8 GiB	100	-	snap-0682a7b...	2024/11/11 15:15 GMT+5:...

Volume ID: vol-04d3eeb483ab7199d

SettingsClose

Details

Status checks

Monitoring

Tags

Volume ID

vol-04d3eeb483ab7199d

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Fast snapshot restored

No

Attached resources

-

Size

4 GiB

Volume state

Available

Availability Zone

us-east-1a

Outposts ARN

-

Type

gp3

IOPS

3000

Created

Mon Nov 11 2024 16:53:12 GMT+0530 (India Standard Time)

Volume status

Okay

Throughput

125

Multi-Attach enabled

No

Detached and deleted old 2gp volume

Volumes (3)

Info

Actions

Create volume

Search

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	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	
5	gp3	2 GiB	3000	125	-	2024/11/11 15:29 GMT+5:...	us-east-1a	Available	N
0d	gp3	4 GiB	3000	125	snap-0c593d8...	2024/11/11 16:53 GMT+5:...	us-east-1a	Available	N
e	gp2	8 GiB	100	-	snap-0682a7b...	2024/11/11 15:15 GMT+5:...	us-east-1a	In-use	N

Successfully deleted volume vol-0b09d0ef176cab56.

Volumes (1/2)

Info

Search

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Actions

Create volume

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
	snapshot	vol-04d3eeb483ab7199d	gp3	4 GiB	3000	125	snap-0c593d8...	2024/11/11 16:53 GMT+5:..
	root volume	vol-0b6801bafcd60f06e	gp2	8 GiB	100	-	snap-0682a7b...	2024/11/11 15:15 GMT+5:..

Again attached 4gp snapshot volume to the instance

Successfully attached volume `vol-04d3eeb483ab7199d` to instance `i-06ecf5d9fff20cbde`.

Volumes (1/2) Info

Search

Refresh

Actions

Create volume

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Settings

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
☒	snapshot	<code>vol-04d3eeb483ab7199d</code>	gp3	4 GiB	3000	125	<code>snap-0c593d8...</code>	2024/11/11 16:53 GMT+5:...
☐	root volume	<code>vol-0b6801bafcd60f06e</code>	gp2	8 GiB	100	-	<code>snap-0682a7b...</code>	2024/11/11 15:15 GMT+5:...

Volume ID: `vol-04d3eeb483ab7199d` (snapshot)

Details

Status checks

Monitoring

Tags

Volume ID

`vol-04d3eeb483ab7199d` (snapshot)

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Fast snapshot restored

No

Size

4 GiB

Volume state

In-use

Availability Zone

us-east-1a

Type

gp3

IOPS

3000

Created

Mon Nov 11 2024 16:53:12 GMT+0530 (India Standard Time)

Volume status

Insufficient data

Throughput

125

Multi-Attach enabled

No

Snapshot EBS volume mounted

```
[ec2-user@ip-172-31-30-251 ~]$ sudo mkdir /data5
[ec2-user@ip-172-31-30-251 ~]$ sudo mount /dev/xvdf /data5
[ec2-user@ip-172-31-30-251 ~]$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
xvda	202:0	0	8G	0	disk	
xvda1	202:1	0	8G	0	part	/
xvdf	202:80	0	4G	0	disk	/data5

```
[ec2-user@ip-172-31-30-251 ~]$
```